

EXP-1 Image Handling and Pixel Transformations Using OpenCV

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Step1: Load an image from your local directory and display it.

```
In [40]: import cv2  
import matplotlib.pyplot as plt
```

```
In [41]: # Read the image using OpenCV  
img = cv2.imread("C:/Users/acer/Downloads/a-cute-puppy-lying-among-flowers-at-sunset-capturing-a-serene-and-joyful-mo")
```

```
In [42]: # Convert BGR (OpenCV's default) to RGB (Matplotlib's expected color order)  
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

```
In [43]: # Display the image using Matplotlib  
plt.imshow(img_rgb, cmap='viridis') # You can change 'viridis' to another cmap or use None for RGB images  
plt.title("Original Image")  
plt.axis('off') # Removes axis ticks and labels  
plt.show()
```

Original Image



Step2: o Draw a line from the top-left to the bottom-right of the image. o Draw a circle at the center of the image. o Draw a rectangle around a specific region of interest in the image. o Add the text "OpenCV Drawing" at the top-left corner of the image. Draw a line from the top-left to the bottom-right of the image

```
In [44]: # Load the image
image = cv2.imread("C:/Users/acer/Downloads/a-cute-puppy-lying-among-flowers-at-sunset-capturing-a-serene-and-joyful-
```

```
In [45]: # Convert BGR (OpenCV's default) to RGB (Matplotlib's expected color order)
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

```
In [46]: img_rgb.shape
```

```
Out[46]: (700, 1246, 3)
```

```
In [47]: # Draw a line from top-left to bottom-right
line_img = cv2.line(img_rgb, (768, 0), (0, 600), (0, 255, 0), 4) # cv2.line(image, start_point, end_point, color, th
```

```
In [48]: plt.imshow(line_img, cmap='viridis')
plt.title("Image with Line")
plt.axis('off')
plt.show()
```

Image with Line



Draw a circle at the center of the image.

```
In [49]: # Load the image
image = cv2.imread("C:/Users/acer/Downloads/a-cute-puppy-lying-among-flowers-at-sunset-capturing-a-serene-and-joyful-

# Convert BGR (OpenCV's default) to RGB (Matplotlib's expected color order)
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

```
In [50]: img_rgb.shape
```

```
Out[50]: (700, 1246, 3)
```

```
In [51]: circle_img = cv2.circle(img_rgb,(160,350), 80, (0,255,0), 5) # cv2.circle(image, center, radius, color, thickness)
```

```
In [52]: plt.imshow(circle_img, cmap='viridis')
plt.title("Image with Circle")
plt.axis('off')
plt.show()
```

Image with Circle



Draw a rectangle around the whole image

```
In [53]: # Load the image
img = cv2.imread("C:/Users/acer/Downloads/a-cute-puppy-lying-among-flowers-at-sunset-capturing-a-serene-and-joyful-mo

# Convert BGR (OpenCV's default) to RGB (Matplotlib's expected color order)
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

```
In [54]: img.shape
```

```
Out[54]: (700, 1246, 3)
```

```
In [55]: # Draw a rectangle around the whole image
rectangle_img = cv2.rectangle(img_rgb, (0, 0), (768, 600), (0, 0, 255), 10) # cv2.rectangle(image, start_point, end_
```

```
In [56]: plt.imshow(rectangle_img, cmap='viridis')
plt.title("Image with Rectangle")
plt.axis('off')
plt.show()
```

Image with Rectangle



Add the text "Puppy" at the top-left corner of the image.

```
In [57]: import cv2
import matplotlib.pyplot as plt

# Load image fresh
img = cv2.imread("C:/Users/acer/Downloads/a-cute-puppy-lying-among-flowers-at-sunset-capturing-a-serene-and-joyful-mo")

# Convert BGR to RGB
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# Add text
text_img = cv2.putText(img_rgb, "OpenCV Drawing", (10,30),
                        cv2.FONT_HERSHEY_SIMPLEX, 1,
                        (255,255,255), 2)

plt.imshow(text_img)
plt.title("Puppy")
plt.axis("off")
plt.show()
```



Step3: o Convert the image from RGB to HSV and display it. o Convert the image from RGB to GRAY and display it. o Convert the image from RGB to YCrCb and display it. o Convert the HSV image back to RGB and display it.

```
In [58]: # Load the image
image = cv2.imread("C:/Users/acer/Downloads/a-cute-puppy-lying-among-flowers-at-sunset-capturing-a-serene-and-joyful-
```

```
In [59]: image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
```

```
In [60]: # Original RGB Image
plt.imshow(image_rgb)
plt.title("Original RGB Image")
plt.axis("off")
```

```
Out[60]: (np.float64(-0.5), np.float64(1245.5), np.float64(699.5), np.float64(-0.5))
```


Original RGB Image

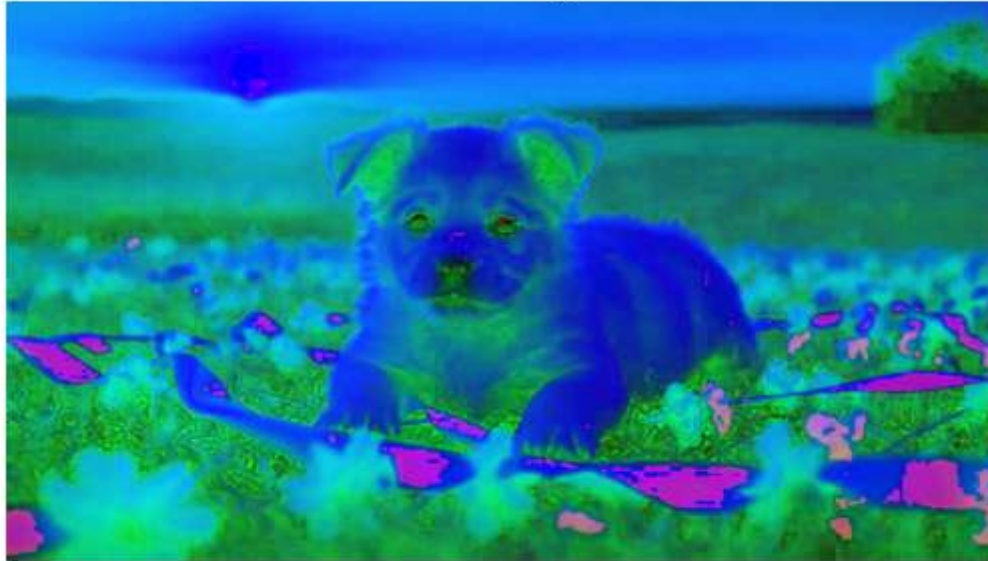


```
In [61]: # Convert RGB to HSV  
image_hsv = cv2.cvtColor(image_rgb, cv2.COLOR_RGB2HSV)
```

```
In [62]: # HSV Image  
plt.imshow(image_hsv)  
plt.title("HSV Image")  
plt.axis("off")
```

```
Out[62]: (np.float64(-0.5), np.float64(1245.5), np.float64(699.5), np.float64(-0.5))
```

HSV Image



```
In [63]: # Convert RGB to GRAY  
image_gray = cv2.cvtColor(image_rgb, cv2.COLOR_RGB2GRAY)
```

```
In [64]: # Grayscale Image  
plt.imshow(image_gray, cmap='gray')  
plt.title("Grayscale Image")  
plt.axis("off")
```

```
Out[64]: (np.float64(-0.5), np.float64(1245.5), np.float64(699.5), np.float64(-0.5))
```


Grayscale Image



```
In [65]: # Convert RGB to YCrCb  
image_ycrb = cv2.cvtColor(image_rgb, cv2.COLOR_RGB2YCrCb)
```

```
In [66]: # YCrCb Image  
plt.imshow(image_ycrb)  
plt.title("YCrCb Image")  
plt.axis("off")
```

```
Out[66]: (np.float64(-0.5), np.float64(1245.5), np.float64(699.5), np.float64(-0.5))
```

YCrCb Image



```
In [67]: # Convert HSV back to RGB  
image_hsv_to_rgb = cv2.cvtColor(image_hsv, cv2.COLOR_HSV2RGB)
```

```
In [68]: plt.imshow(image_hsv_to_rgb)  
plt.title("HSV to RGB Image")  
plt.axis("off")
```

```
Out[68]: (np.float64(-0.5), np.float64(1245.5), np.float64(699.5), np.float64(-0.5))
```

HSV to RGB Image



Step4: o Access and print the value of the pixel at coordinates (100, 100). o Modify the color of the pixel at (200, 200) to white.

```
In [69]: # Modify a block of pixels (300x300) to white, starting from (200, 200)
image[200:500, 200:500] = [255, 255, 255] # Rows: 200-499, Columns: 200-499
```

```
In [70]: # Convert BGR to RGB for displaying with Matplotlib
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
```

```
In [71]: # Display the modified image
plt.imshow(image_rgb)
plt.title("Image with 300x300 White Block")
plt.axis("off")
plt.show()
```

Image with 300x300 White Block



Step5: o Resize the original image to half its size and display it.

```
In [72]: # Load the image  
image = cv2.imread("C:/Users/acer/Downloads/a-cute-puppy-lying-among-flowers-at-sunset-capturing-a-serene-and-joyful-
```

```
In [73]: image.shape
```

```
Out[73]: (700, 1246, 3)
```

```
In [74]: # Resize the image to half its size  
resized_image = cv2.resize(image, (768 // 2, 600 // 2)) # (new_width, new_height)
```

```
In [75]: # Convert BGR to RGB for displaying with Matplotlib  
resized_image_rgb = cv2.cvtColor(resized_image, cv2.COLOR_BGR2RGB)
```

```
In [76]: resized_image_rgb.shape
```

```
Out[76]: (300, 384, 3)
```

```
In [77]: # Display the resized image  
plt.imshow(resized_image_rgb)  
plt.title("Resized Image (Half Size)")
```

```
plt.axis("off")  
plt.show()
```

Resized Image (Half Size)



Step6: o Crop a region of interest (ROI) from the image (e.g., a 100x100 pixel area starting at (50, 50)) and display it.

```
In [78]: # Load the image  
image = cv2.imread("C:/Users/acer/Downloads/a-cute-puppy-lying-among-flowers-at-sunset-capturing-a-serene-and-joyful-
```

```
In [42]: image.shape
```

```
Out[42]: (700, 1246, 3)
```

```
In [43]: # Crop a 300x300 region starting from (50, 50)  
roi = image[50:350, 50:350] # Rows: 50-349, Columns: 50-349
```

```
In [44]: # Convert BGR to RGB for displaying with Matplotlib
roi_rgb = cv2.cvtColor(roi, cv2.COLOR_BGR2RGB)
```

```
In [45]: # Display the cropped region (ROI)
plt.imshow(roi_rgb)
plt.title("Cropped Region of Interest (ROI)")
plt.axis("off")
plt.show()
```

Cropped Region of Interest (ROI)



Step7: o Flip the original image horizontally and display it. o Flip the original image vertically and display it.

```
In [46]: # Load the image
image = cv2.imread("C:/Users/acer/Downloads/a-cute-puppy-lying-among-flowers-at-sunset-capturing-a-serene-and-joyful-
```

```
In [47]: # Flip the image horizontally (left-right)
flipped_horizontally = cv2.flip(image, 1)
```



```
In [48]: # Convert BGR to RGB for displaying with Matplotlib
flipped_horizontally_rgb = cv2.cvtColor(flipped_horizontally, cv2.COLOR_BGR2RGB)
```

```
In [49]: # Horizontal flip
plt.imshow(flipped_horizontally_rgb)
plt.title("Flipped Horizontally")
plt.axis("off")
```

```
Out[49]: (np.float64(-0.5), np.float64(1245.5), np.float64(699.5), np.float64(-0.5))
```

Flipped Horizontally



```
In [50]: # Flip the image vertically (up-down)
flipped_vertically = cv2.flip(image, 0)
```

```
In [51]: # Convert BGR to RGB for displaying with Matplotlib
flipped_vertically_rgb = cv2.cvtColor(flipped_vertically, cv2.COLOR_BGR2RGB)
```

```
In [52]: # Vertical flip
plt.imshow(flipped_vertically_rgb)
plt.title("Flipped Vertically")
plt.axis("off")
```

```
Out[52]: (np.float64(-0.5), np.float64(1245.5), np.float64(699.5), np.float64(-0.5))
```

Flipped Vertically



Step8: o Save the final modified image to your local directory. Image with flag

```
In [28]: import cv2
import matplotlib.pyplot as plt

img = cv2.imread("C:/Users/acer/Downloads/a-cute-puppy-lying-among-flowers-at-sunset-capturing-a-serene-and-joyful-mo")

flag = cv2.imread("C:/Users/acer/Downloads/Flag_of_India.png", cv2.IMREAD_UNCHANGED)

flag = cv2.resize(flag, (50, 30))

# Flag position change here
x = 650 # left-right position
y = 400 # up-down position

b, g, r, a = cv2.split(flag)

flag_rgb = cv2.merge((b, g, r))

mask = a / 255.0
```

```
for c in range(3):  
    img[y:y+30, x:x+50, c] = (1-mask) * img[y:y+30, x:x+50, c] + mask * flag_rgb[:, :, c]  
  
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)  
  
plt.figure(figsize=(8,6))  
plt.imshow(img)  
plt.axis("off")  
plt.show()
```



In []:

In []: